

RKL-01 Submersible Liquid Level Transmitter



RKL-01 Submersible Liquid Level Transmitter adopts temperature-compensating isolating-membrane diffusing silicon pressure sensor as a sensing element.supported with the special cable provided with air venting conduit as well as special water-tightness technology, which ensure water tightness of immersing-type sensor and make the reference pressure chamber connecting with the ambient atmosphere so as to attain accurate results of measurement and excellent stability.

FEATURES

- High accuracy
- High sensitivity
- Good stability
- All stainless steel structure probe
- Anti-corrosion material optional
- Easy Installation

SPECIFICATIONS

Item	Technical specifications
Range	0 ~ 0.5m...200mH2O or 0 ~ 5KPa...2MPa
Output	4-20mA,0-5V,0-10V,RS485
Supply Voltage	5V,10-30VDC,24V typ.
Over Pressure	2×FS
Measuring Medium	The liquid(not sticky) compatible with 316 stainless steel
Total Accuracy	0.1%FS,0.3%FS,0.25%FS,0.5%FS
Long-term Stability	0.1%FS/year typ.,0.2%FS/year max.
Ingress Protection	IP68
Operating Temperature	-40℃~ +80℃
Compensating Temperature	-10℃~ 70℃
Temperature Drifting	0.03%FS/℃ typ.,0.05%FS/℃
Cycle Life	1*10 ⁸ @25℃
Main Material	Sensor:316L,housing:304,cable:PUR, other materials can be customized
Power Consumption	Current output:(U*0.02)W, Voltage output:(U*0.008)W, Digital output:(U*0.015)W
Load Capacity	Current output:≤(U-7)/0.02Ω, Voltage output:≥100kΩ
Weight(unpacked)	Approx. 230g
Storage Condition	10℃-50℃@20%-90%RH

OUTPUT CHARACTERISTICS

● Current

Characteristic transfer function:

$$H=(I-4)/(20-4)*\text{Scale range}$$

(where H= water level(m),I = output current(mA))

● Voltage

Characteristic transfer function:

$$H=U/(\text{full scale voltage-zero point voltage})*\text{Scale range}$$

(where H = water level(m),U = output voltage(V))

● RS485

If the transmission distance is over 100m, please add one 120Ω terminal matching resistance on the front end and back end of bus interface respectively.See appendix for communication protocol.

MOUNTING

When the sensor is installed in static water such as in pools, water towers, probe is immersed into the bottom and should be as far as possible away from the pump or valves. The terminal box should keep above water surface and prevent water penetrating into cables. Please make sure the airway not be blocked.

When the sensor is installed in dynamic water such as dams or rivers, probe should be inserted into a steel pipe (inner dia is around 45mm), burrowing several holes at different height on the pipe side wall which is opposite to the water flow direction. The terminal box should keep above water surface and prevent water penetrating into cables. Please make sure the airway not be blocked.

WARRANTY

This product is warranted to be free of defects in materials and construction for a period of 12 months from date of lead time.

Liability is limited to repair or replacement of defective item.

ELECTRICAL CONNECTIONS

Connector (cable)	Current	RS485	Voltage
Red	V+	V+	V+
Yellow		RS485A	Signal
Blue		RS485B	
Green	Signal+	V-	V-

Note: This product has been tested and complies with European CE requirements for EMC directive.

Communication Protocol (MODBUS)

Transmission mode: MODBUS-RTU, Baud rate: 9600bps, Data bits: 8, Stop bit: 1, Check bit: no

Slave address: the factory default is 01H (set according to the need, 00H to FFH)

- The 03H Function Code Example: Read The Level Value

Host Scan Order (slave address: 0x01)

01 03 00 04 00 01 C5CB

Slave Response

01 03 02 01 B4 B9A3

Level(m): (01B4)H = (436)D, $436/100 = 4.36\text{m}$

- The 06H Function Code Example: Modify the slave address (Changed the 01H to 02H)

Step 1:

Host Scan Order (Changed the 01H to 02H): (ensure that no other devices on the bus at this time)

01 06 00 00 00 02 080B

Slave Response:

01 06 00 00 00 02 080B

Step 2:

Host Scan Order (Save the new address):

02 06 00 0F 00 00 B9FA

Slave Response:

02 06 00 0F 00 00 B9FA

Note:

- All underlined is fixed bit;
- The last two bytes is CRC check command.